Repeating Efficiently

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 10: a condition can decide whether an action runs.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Trace a loop by testing its condition before every repetition. Try the recall before opening notes.

Your learning target

I can trace a **loop** by testing its condition before every repetition.

I can distinguish the number of repetitions from the final value and include zero repetitions.

I can explain why an update reaches a **stopping condition** or why a faulty **loop** keeps repeating.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

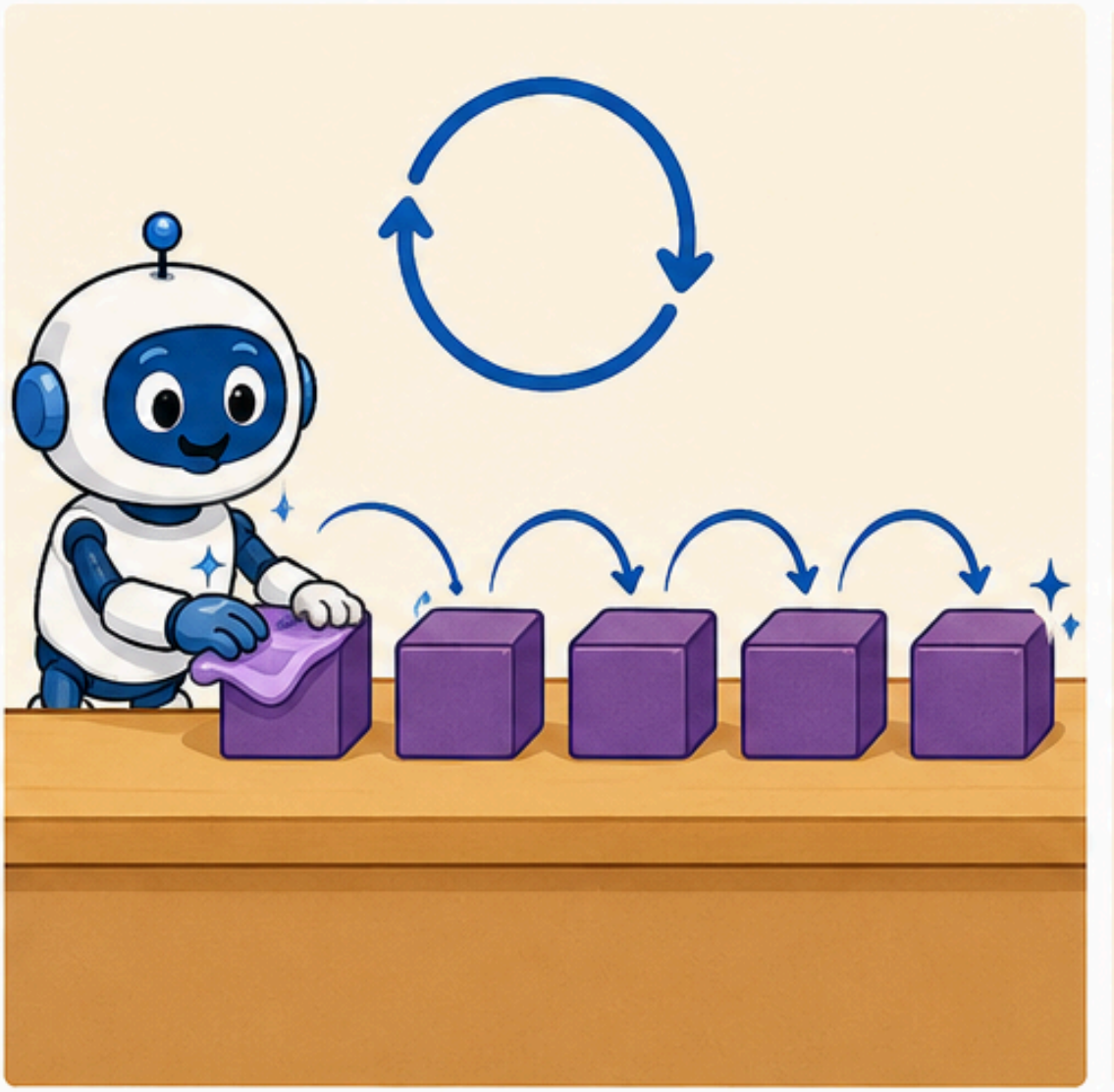
Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A **loop** repeats a useful step until a **stopping condition** is met.



Your target

LEARNING TARGET A robot stamps the next unstamped card until it has stamped five cards. Only three cards are available. Write a stopping rule that handles both limits and trace the number stamped.

Learner name:

Readiness for the connected explanation

Score is 4. If score is below 5, add 3; otherwise subtract 2. Report and stop. Does reaching 7 cause another test? Explain.

This week's end goal

By the end, I can check a **loop** condition before each update and count **iteration**s.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Repeating Efficiently case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY Instead of writing draw-side and turn four times, a square algorithm can repeat those steps four times.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

A paper counter starts at 1. While the counter is below 7, add 2 to it. When the test is false, report the counter and stop.

The word while means test first, run the **body** only if true, then return to the test using the updated value. The **body** here is just add 2.

The counter follows 1, 3, 5, 7. At 1, 3 and 5 the condition is true, so each test is followed by one addition. At 7 the condition is false. No addition follows that final test. The **loop** has three **iteration**s and reports 7.

Compare the Week 10 instruction if counter is below 7, add 2, then report. From 1 that single decision reports 3. A while **loop** explicitly returns to its test; an if decision does not.

Draw a pointer to the instruction being executed instead of deciding for yourself when to repeat.

STUDENT LESSON · 3 OF 18

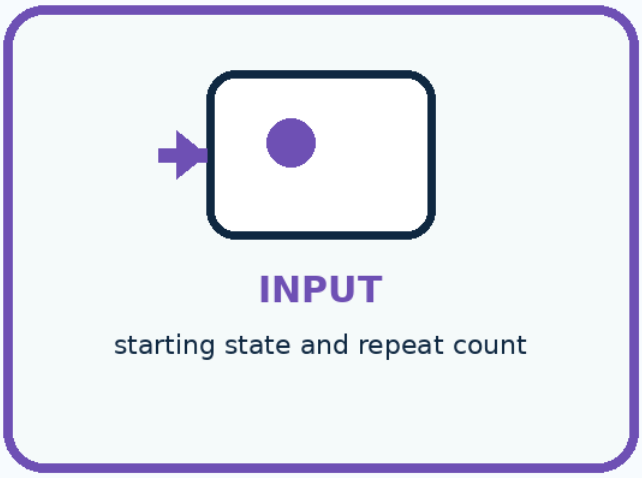
See how it works

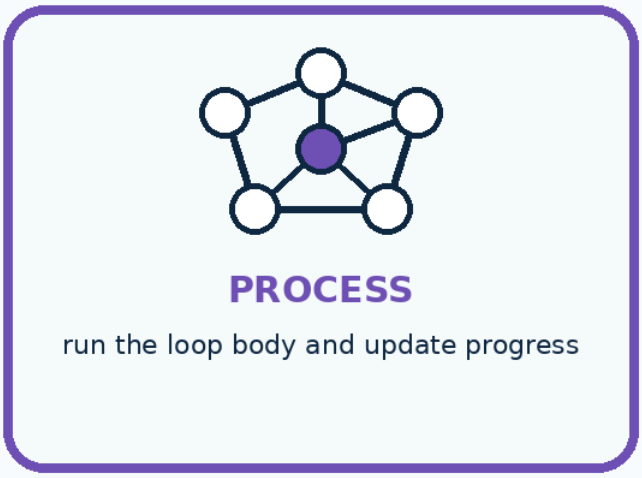
> Follow the mechanism for Repeating Efficiently. Connect each labelled input, deciding step and result.

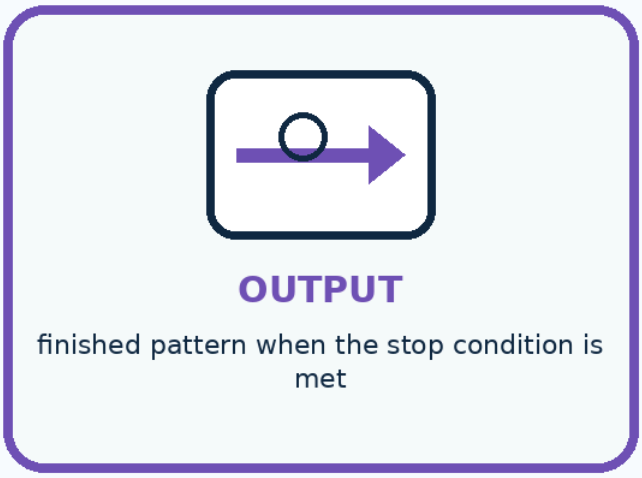
VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT starting state and repeat count

PROCESS run the **loop body** and update **progress**

OUTPUT finished pattern when the stop condition is met

Key idea

CORE IDEA A **loop** repeats a useful step until a **stopping condition** is met.

How the mechanism works

UNDER THE HOOD A **loop** requires an initialization, **body**, update, and termination condition.

Repetition is safe only when each **iteration** moves state toward stopping; otherwise the **loop** may run forever or the wrong number of times.

Draw a fresh example with all three stages:

A closer explanation

A complete loop trace

COUNTER BEFORE TEST	IS COUNTER BELOW 7	NEXT ACTION
1	True	Add 2 to reach 3
3	True	Add 2 to reach 5
5	True	Add 2 to reach 7
7	False	Report 7 and stop

Read the rows from top to bottom as one run. Each true test performs one addition and returns to the test. The final false test performs no addition.

The trace has four tests but only three **iteration**s. The false test is **evidence** that this run stops. A list ending at 5 would be incomplete, while a list continuing to 9 would include an addition the **loop** did not authorise.

Write the current value, the test result and the action on every row. Repeated values can matter: if the **body** does not change the state used by the test, the same true test may occur again.

Do not erase these rows simply because they look unhelpful.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

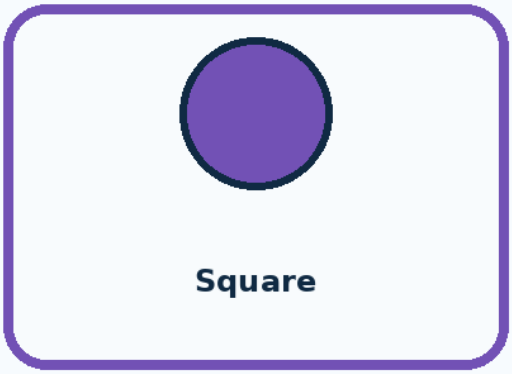
Compare the cases

> Compare the supplied cases for Repeating Efficiently. Identify the detail that changes the conclusion.

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Square Clue: repeat move and turn 4 times. Meaning: counted **loop**.

The case to investigate

CASE 2 — Pick cards Clue: repeat until pile empty. Meaning: condition **loop**.

The case to investigate

CASE 3 — No counter update Clue: condition never changes. Meaning: infinite **loop** risk.

The case to investigate

CASE 4 — Stop too early Clue: one item remains. Meaning: off-by-one error.

Keep this boundary clear

BOUNDARY Repeat forever is not an efficient algorithm unless endless operation is truly the goal and can be stopped safely.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Start at 10 instead of 2.

Predicted result

Prediction: The **loop** runs zero times and reports 10.

Why the result follows

Reason: Its first condition is already false, so no addition happens.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

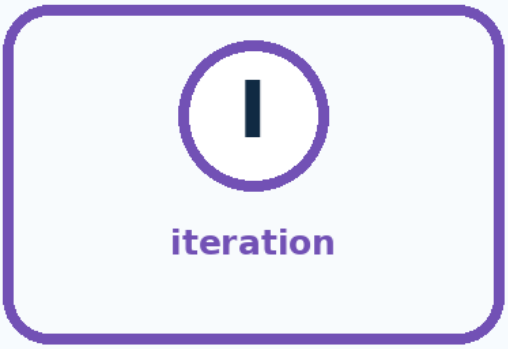
> Use the vocabulary of Repeating Efficiently to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
loop	control structure that repeats steps	
iteration	one pass through the loop	

Word	Meaning	My example
loop body	steps repeated each time	
stopping condition	rule that ends repetition	

Misconception detective

Find the unsupported claim

CLAIM TO REPAIR A **loop** is just copying the same words many times.

Use one vocabulary word and one piece of **evidence** to repair it:

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Body	The instructions performed on each iteration
Progress	A change that brings the state closer to a stopping condition

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STUDENT LESSON · 6 OF 18

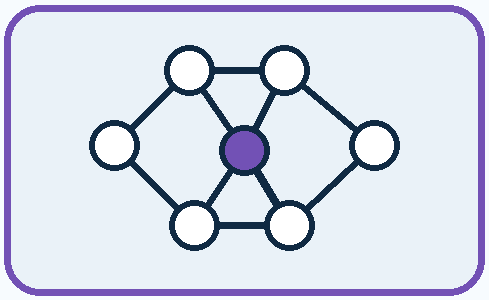
Follow a worked example

> Follow the worked trace for Repeating Efficiently, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE Instead of writing draw-side and turn four times, a square algorithm can repeat those steps four times.

Step 1

STEP 1 — OBSERVE starting state and repeat count

Step 2

STEP 2 — TRACE run the **loop body** and update **progress**

Step 3

STEP 3 — CONCLUDE finished pattern when the stop condition is met

Model answer

MODEL REASONING Place only the repeated steps inside the **loop**, initialize the counter, update it once per pass, and stop after the required count. Trace each **iteration** to catch one-too-many or one-too-few errors.

A claim to avoid

DO NOT OVERCLAIM Repeat forever is not an efficient algorithm unless endless operation is truly the goal and can be stopped safely.

What new **evidence** could change this answer?

A closer explanation

Starting counter	Values at successive tests	Iterations and reported value
5	5 true then 7 false	1 iteration and report 7
6	6 true then 8 false	1 iteration and report 8
7	7 false	0 iterations and report 7
8	8 false	0 iterations and report 8

While tests before the **body**. If the first test is false, the **body** runs zero times. Do not add 2 once just because a **loop** is present. Starting at 7 or 8 is a useful test of this rule.

A **loop** that stops when the counter is no longer below 7 does not promise to finish at exactly 7. Starting at 6 produces 8. Passing the boundary is allowed by these instructions.

Always report the actual final state instead of replacing it with the threshold.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Repeating Efficiently

A fictional paper counter advances through a display until its condition becomes false.

The facts we will use

Supplied facts

Start at 2. While the current counter is below 10, add 3.

Supplied facts

Test before each addition; report when the condition is false.

AI ACADEMY / AI-1/11

Condition checks before additions

	Current	Below 10?	Next
Check 1	2	true	5
Check 2	5	true	8
Check 3	8	true	11
Check 4	11	false	stop

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/11

Repeating Efficiently

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First test

2<10 is true: update to 5.



2. Repeat

5<10: update to 8. 8<10: update to 11.



3. Stop test

11<10 is false; do not add again.



4. Report

Final counter 11 after three additions.

CHANGE ONE THING

Start at 10 instead of 2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. First test	2<10 is true: update to 5.
2. Repeat	5<10: update to 8. 8<10: update to 11.
3. Stop test	11<10 is false; do not add again.
4. Report	Final counter 11 after three additions.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The condition checks whether to do the next update; it does not constrain every resulting value to stay below 10. The final 11 is therefore allowed by this procedure.

Record both the condition and counter to distinguish a correct stop from an assumed endpoint.

What this example does not establish

Keep the stated limit

This **loop** stops because positive increments eventually cross the limit. Other updates can

fail to make **progress**.

Keep the stated limit

Read the labelled picture: Trace a fresh start of 9. How many additions occur, and must the report equal 10?

Keep the stated limit

Change one thing: Start at 10 instead of 2. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Repeating Efficiently claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Square	repeat move and turn 4 times	counted loop
Pick cards	repeat until pile empty	condition loop
No counter update	condition never changes	infinite loop risk

Case	Evidence	What it shows
Stop too early	one item remains	off-by-one error

Repair two tempting claims

Challenge claim A

CLAIM A A **loop** is just copying the same words many times.

Repair:

Challenge claim B

CLAIM B The **stopping condition** will become true by itself.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

A different mistake says while counter is not equal to 7, add 2. From 6 the values are 6, 8, 10 and then larger even numbers. None is 7, so this equality test never becomes false.

A growing number does not guarantee termination: the update must be able to reach a state that makes this particular test false.

Our paper examples use known whole numbers without a maximum size. For this lesson, an unreadable starting counter means pause and request the value. It does not mean start at zero.

Later programs will need explicit rules for unavailable **input**s and resource limits.

> **A tester runs the below 7 loop only from starting counter 1. Choose one additional start that checks zero repetitions and one that checks passing the boundary without landing on it. Give the expected reported values**

STUDENT LESSON · 8 OF 18

Practise together

> Practise Repeating Efficiently with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A **loop** repeats a useful step until a **stopping condition** is met.

Use a fresh example to explain **loop**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: The **stopping condition** will become true by itself.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

For our known whole-number counters below 7, adding 2 makes the counter larger each time. After enough additions it is at least 7, so the test becomes false.

This explains why this **loop** stops for these **input**s; one successful trace alone would not establish that for every possible procedure.

Now change the **body** to add 0. Starting at 1 gives the same value 1 at every test. Below 7 remains true, and the **loop** never reaches its reporting instruction.

A classroom trace should be stopped by the teacher after the repeated state is established. That intervention is not a successful program result.

Compare the original add 2 loop from starting counter 3 with the faulty add 0 version. Record enough test rows to justify each conclusion. Explain the difference between the final reported value and a value merely seen during a non stopping run

Practise with less help: Check the first test

Use workbook W2 for the connected example “**Loop**s and stopping rules”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Loop**s and stopping rules. Proposed explanation: “Every **loop** runs at least once”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Trace a fresh start of 9. How many additions occur, and must the report equal 10?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

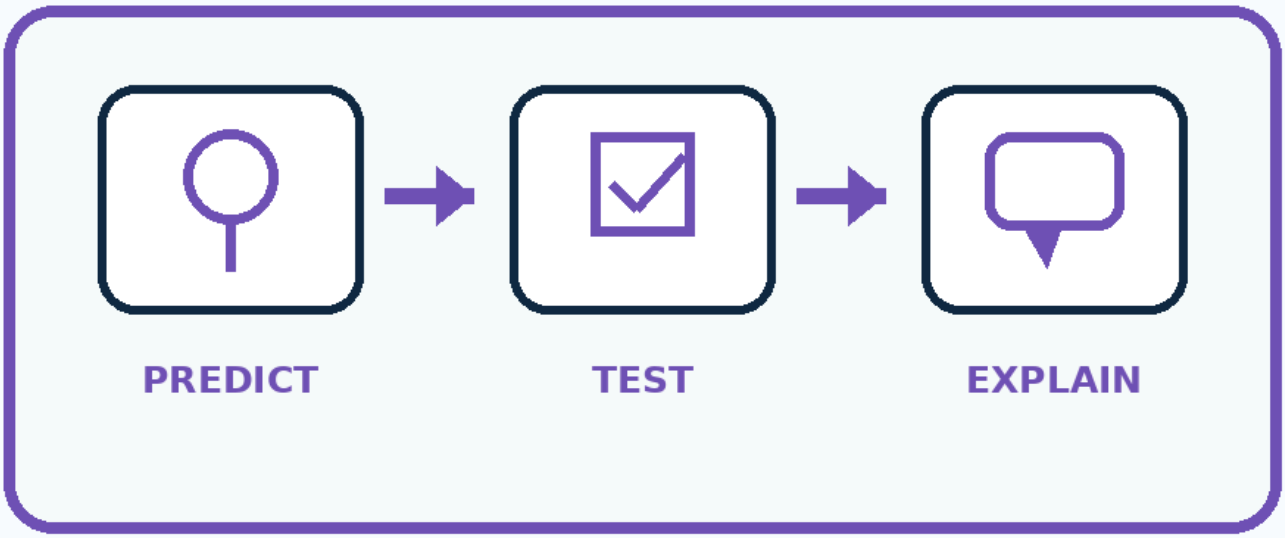
Run the investigation

> Investigate Repeating Efficiently. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Loop Card Drawing

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Compress a repeated drawing algorithm using a **loop** card.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

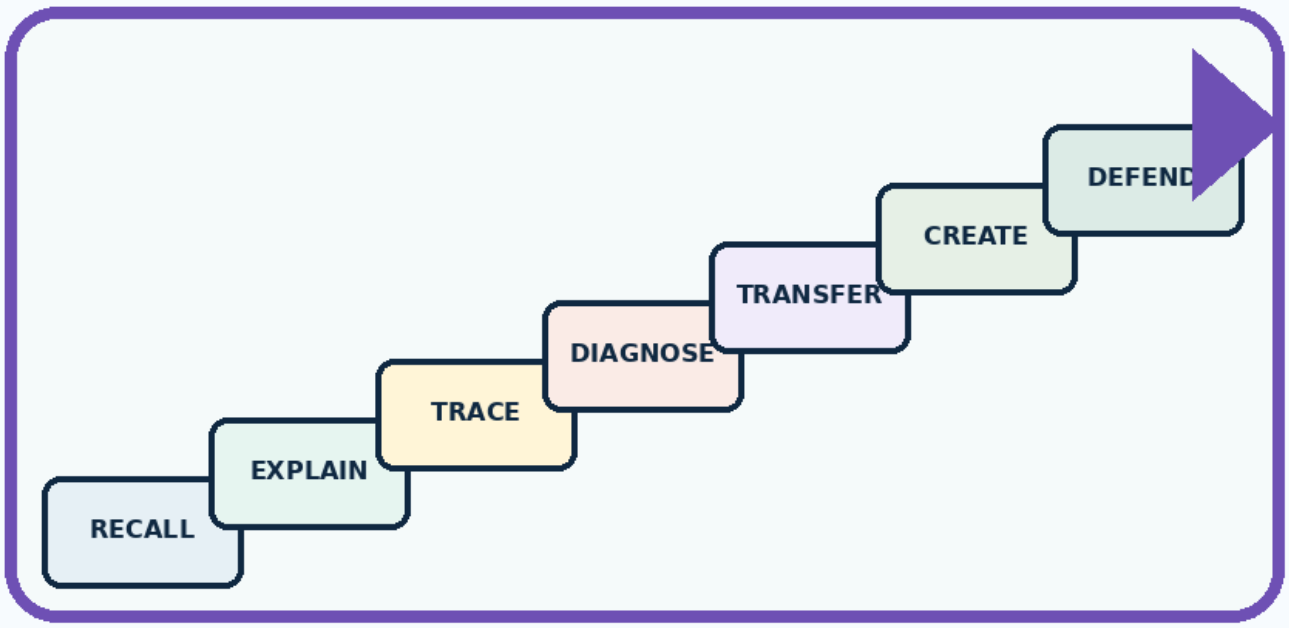
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Repeating Efficiently. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **loop**.

EXPLAIN A **loop** repeats a useful step until a **stopping condition** is met.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: A **loop** is just copying the same words many times.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

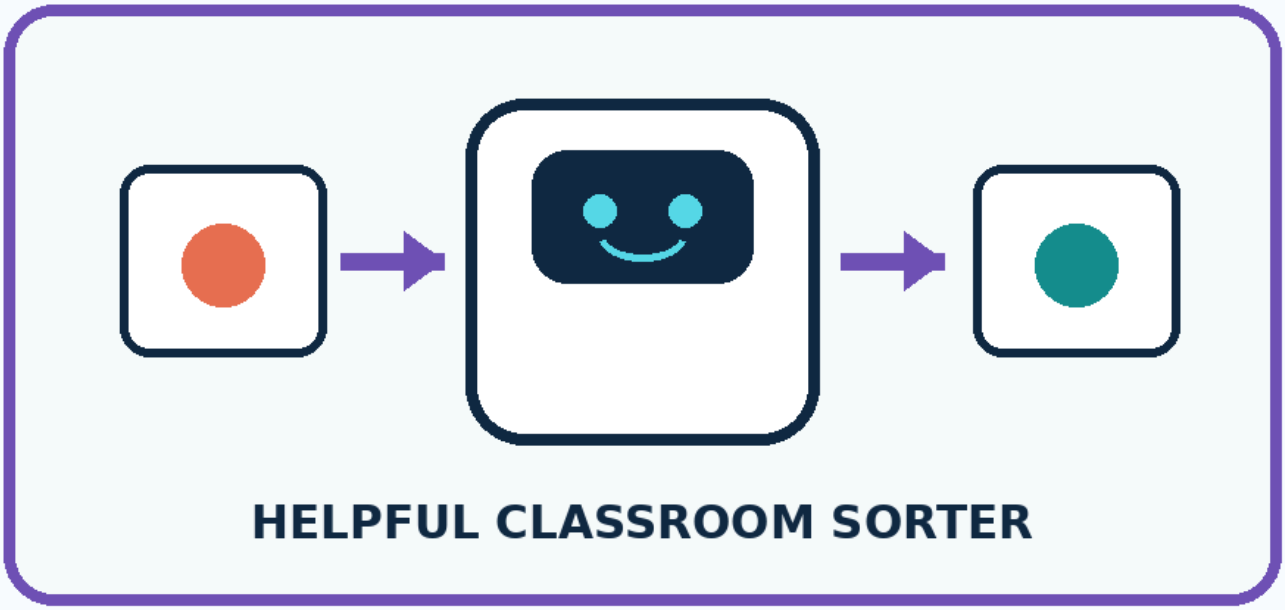
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Use a **loop** to **process** each item card in the test pile, with an explicit stop when no cards remain.

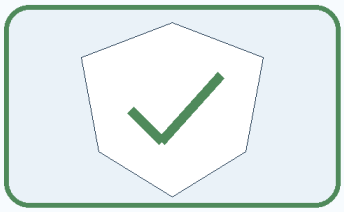
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN Repeat forever is not an efficient algorithm unless endless operation is truly the goal and can be stopped safely.

My project decision:

Evidence for your project

Give any repeated sorter step an explicit **stopping condition** and trace a zero-**iteration** example.

STUDENT LESSON · 12 OF 18

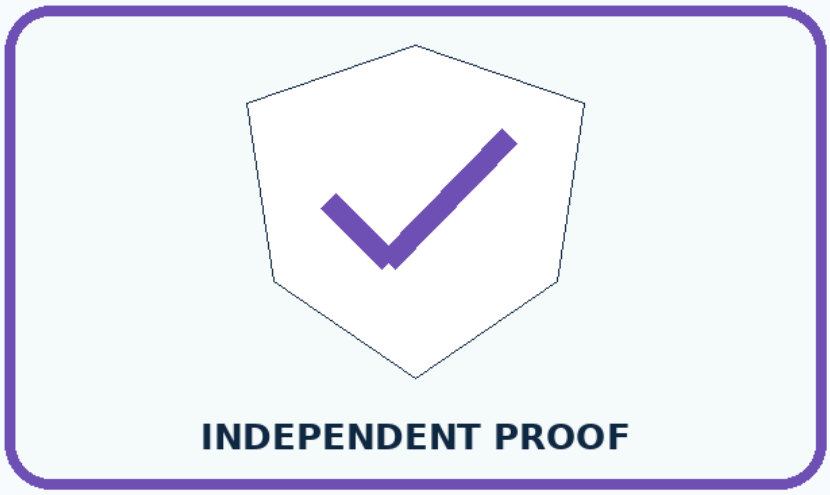
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Compress a repeated drawing into a **loop**, predict the **iteration** count, and explain what makes it stop.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Place only the repeated steps inside the **loop**, initialize the counter, update it once per pass, and stop after the required count. Trace each **iteration** to catch one-too-many or one-too-few errors.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Debug the Cause.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Repeating Efficiently case for supported practice. Repeated examples do not become fresh assessments.

Week 11 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Repeat take the next unstamped card and stamp it four times.

The question

How many cards are stamped? What assumption matters?

Worked explanation

Four, provided four unstamped cards exist and each repeat takes the next one. A limit or end-of-stack check prevents continuing indefinitely.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Repeat with a clear stopping point

Evidence or stage	Value or result
Before	0 stamped
Repeat 1	1 stamped
Repeat 2	2 stamped
Repeat 3	3 stamped
Repeat 4	4 stamped, stop

Repeat take the next unstamped card and stamp it four times.

> How many cards are stamped? What assumption matters?

Four, provided four unstamped cards exist and each repeat takes the next one. A limit or end-of-stack check prevents continuing indefinitely.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Transfer to a countdown

A paper countdown starts at 7. While remaining is above 1, subtract 2. When false, report and stop. Trace the run and count **iteration**s. Repeat from 6 and from 1. Explain why the final value need not equal 1.

What happens from 7 if subtract 2 is replaced by subtract 0?

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 11 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> Repeat clap then tap three times. Write the complete sequence. How many actions occur? What would happen if the stopping instruction were missing?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Use a **loop** to **process** each item card in the test pile, with an explicit stop when no cards remain.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A robot stamps the next unstamped card until it has stamped five cards. Only three cards are available. Write a stopping rule that handles both limits and trace the number stamped.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Repeating Efficiently, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional paper counter advances through a display until its condition becomes false.

By the end, I can	How I will show it
Trace a loop by testing its condition before every repetition	Explain the deciding step, show a trace or test, and state one limit.
Distinguish the number of repetitions from the final value and include zero repetitions	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Explain why an update reaches a stopping condition or why a faulty loop keeps repeating	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Repeating Efficiently without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Trace a **loop** by testing its condition before every repetition

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional paper counter advances through a display until its condition becomes false.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Final counter 11 after three additions.

1. Report

2. First test

3. Repeat

4. Stop test

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report Final counter 11 after three additions. The condition checks whether to do the next update; it does not constrain every resulting value to stay below 10.

The final 11 is therefore allowed by this procedure. Record both the condition and counter to distinguish a correct stop from an assumed endpoint.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Start at 10 instead of 2. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The **loop** runs zero times and reports 10.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The **loop** runs zero times and reports 10. Its first condition is already false, so no addition happens.

Repair a missing building block

Review If/Then Decisions (ai-1/10). Goal: Evaluate a condition using the current state and trace only the selected branch. Short practice: Which **input** detects the difference between the two written conditions?

Explain its entire run.. Return to this week and explain what you changed.

Review Features: Useful Clues (ai-1/8). Goal: Choose recorded properties that preserve distinctions needed for a stated task.

Short practice: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.. Return to this week and explain what you changed.

Review Pictures Become Numbers (ai-1/5). Goal: Encode and reconstruct a small image using an explicit grid convention.

Short practice: Find a position where the original badges differ, then read the changed A back as a picture.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

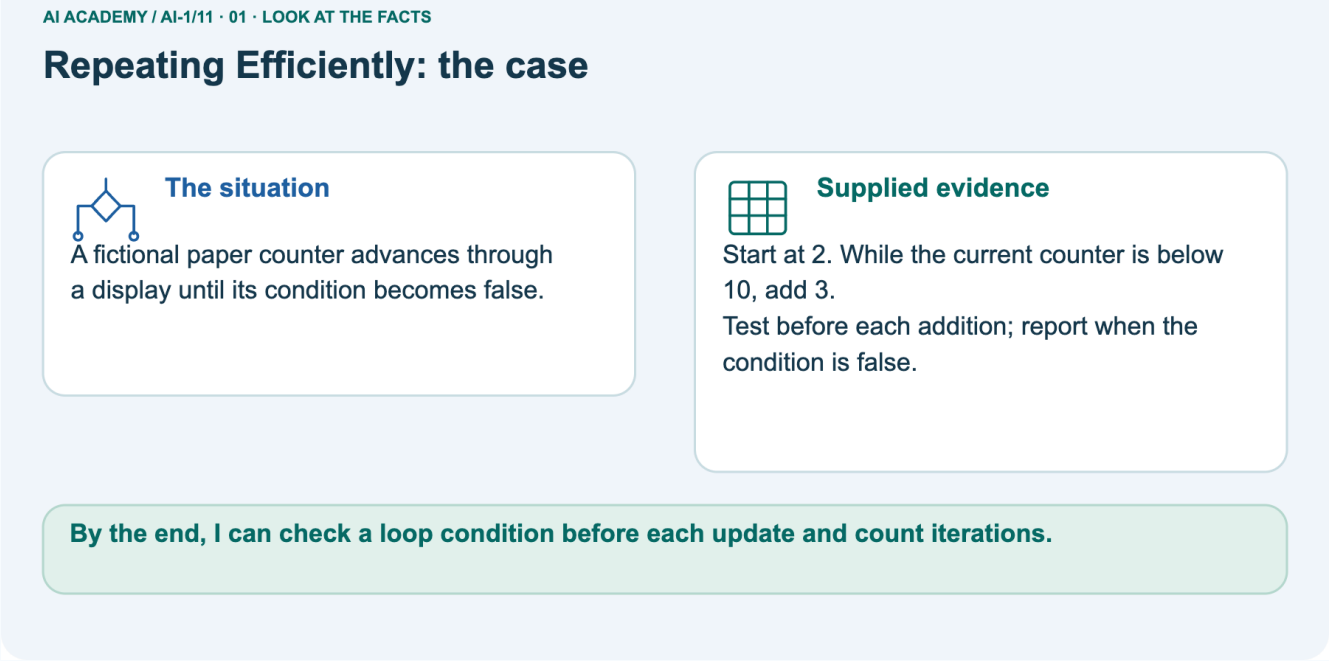
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Repeating Efficiently graphic using its labels, deciding evidence and relevant limitation.

01. Repeating Efficiently: the case

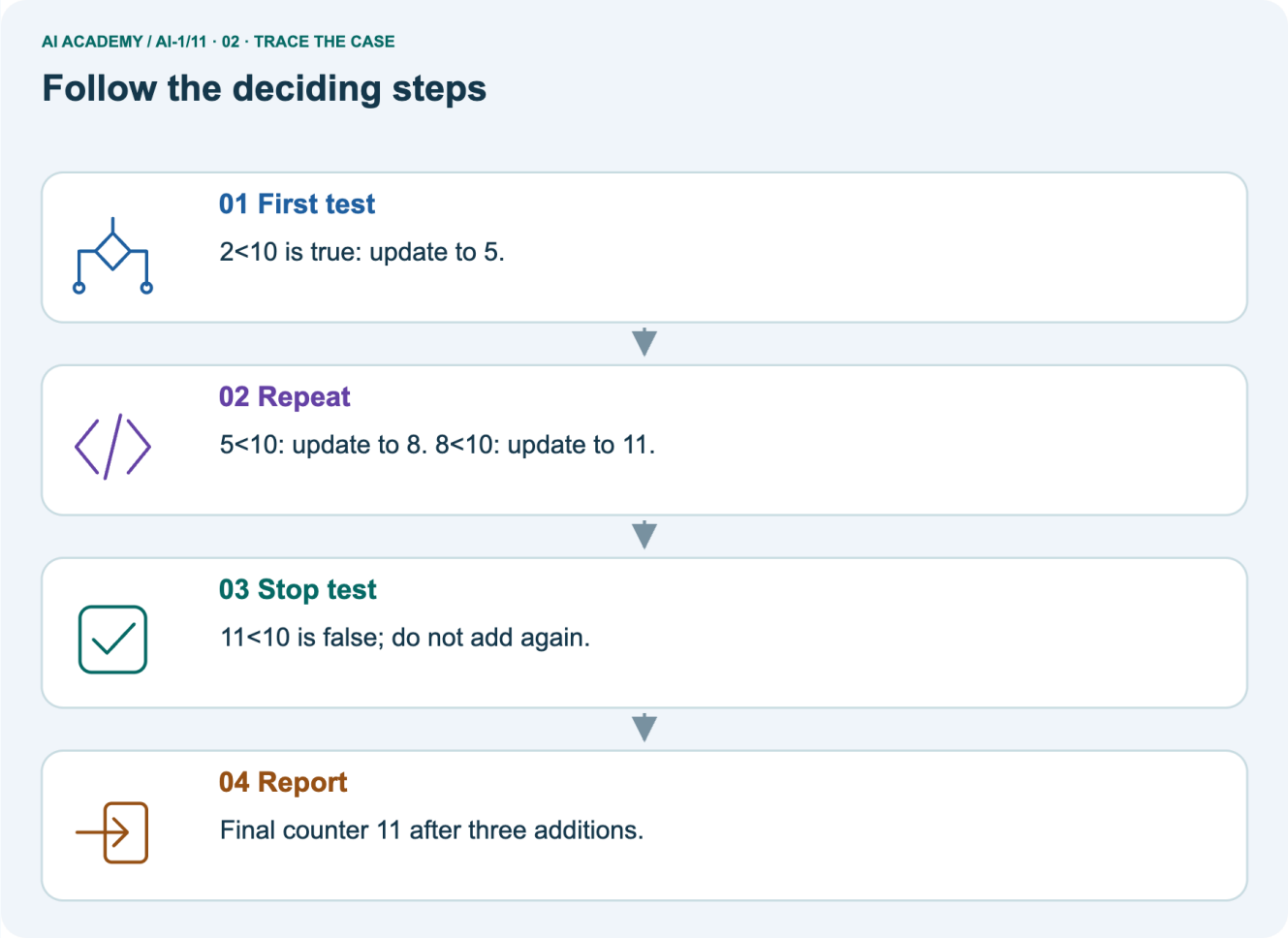


Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional paper counter advances through a display until its condition becomes false. Supplied facts: Start at 2. While the current counter is below 10, add 3. Test before each addition; report when the condition is false.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

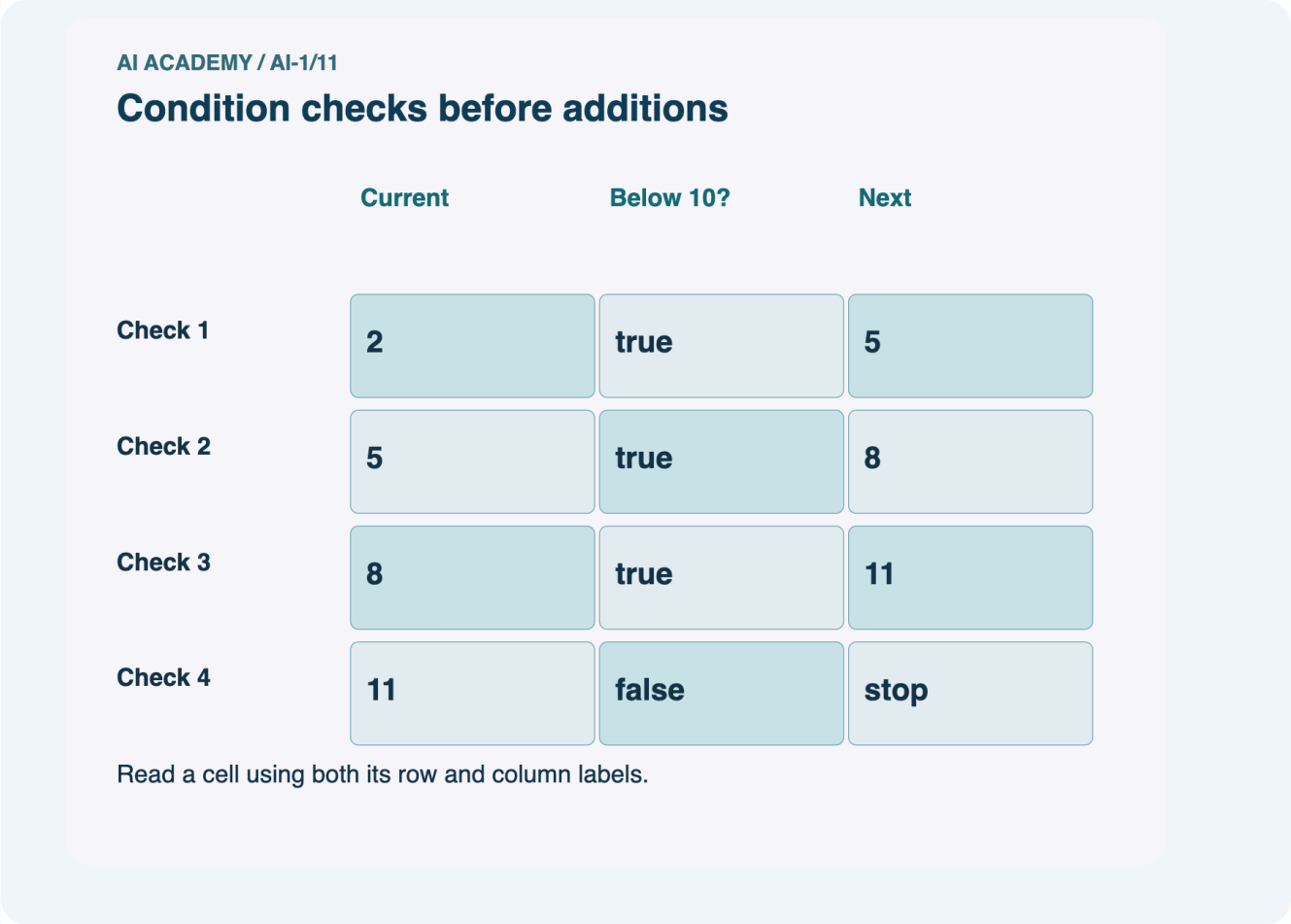


Follow the labelled picture with the matching explanation. Open the vector diagram

First test: 2<10 is true: update to 5. → Repeat: 5<10: update to 8. 8<10: update to 11. → Stop test: 11<10 is false; do not add again. → Report: Final counter 11 after three additions.

Point to each numbered stage and explain which recorded **input** it uses.

03. Condition checks before additions



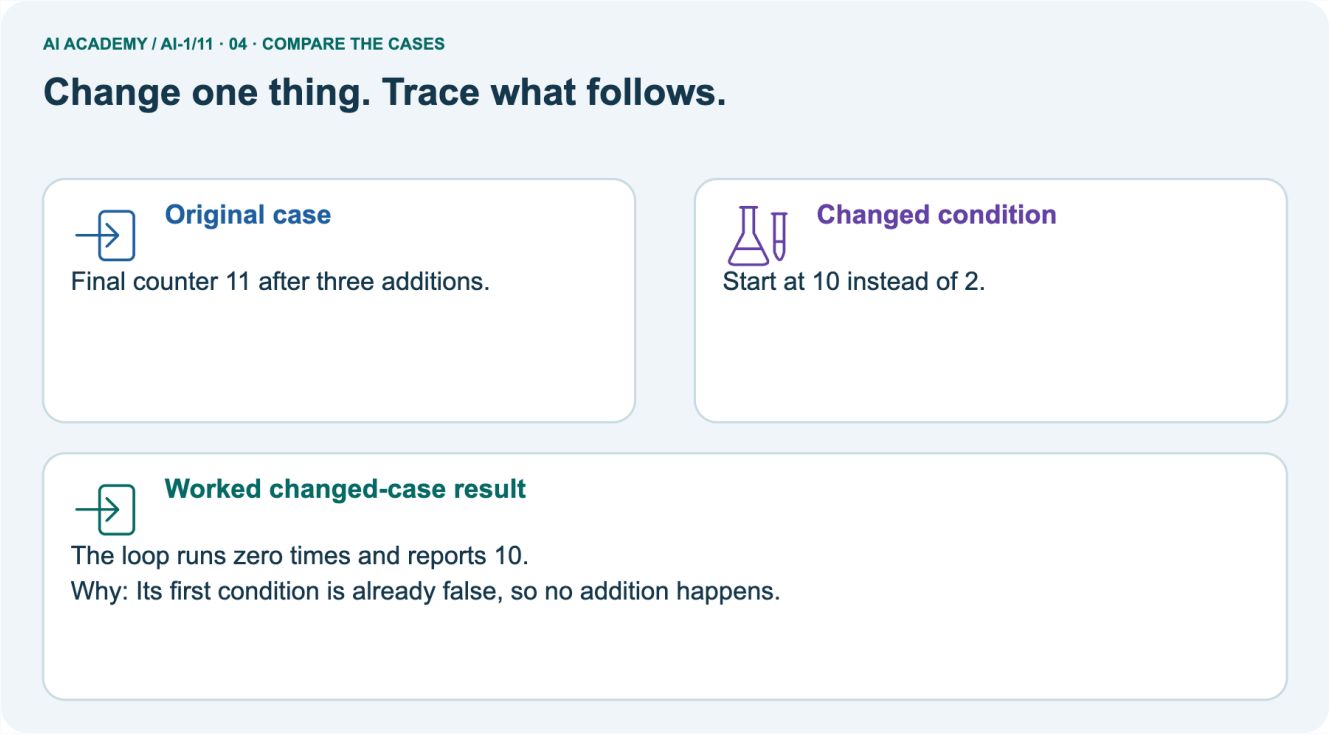
Follow the labelled picture with the matching explanation. Open the vector diagram

Condition checks before additions. Check 1: Current = 2, Below 10? = true, Next = 5. Check 2: Current = 5, Below 10? = true, Next = 8. Check 3: Current = 8, Below 10? = true, Next = 11.

Check 4: Current = 11, Below 10? = false, Next = stop.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



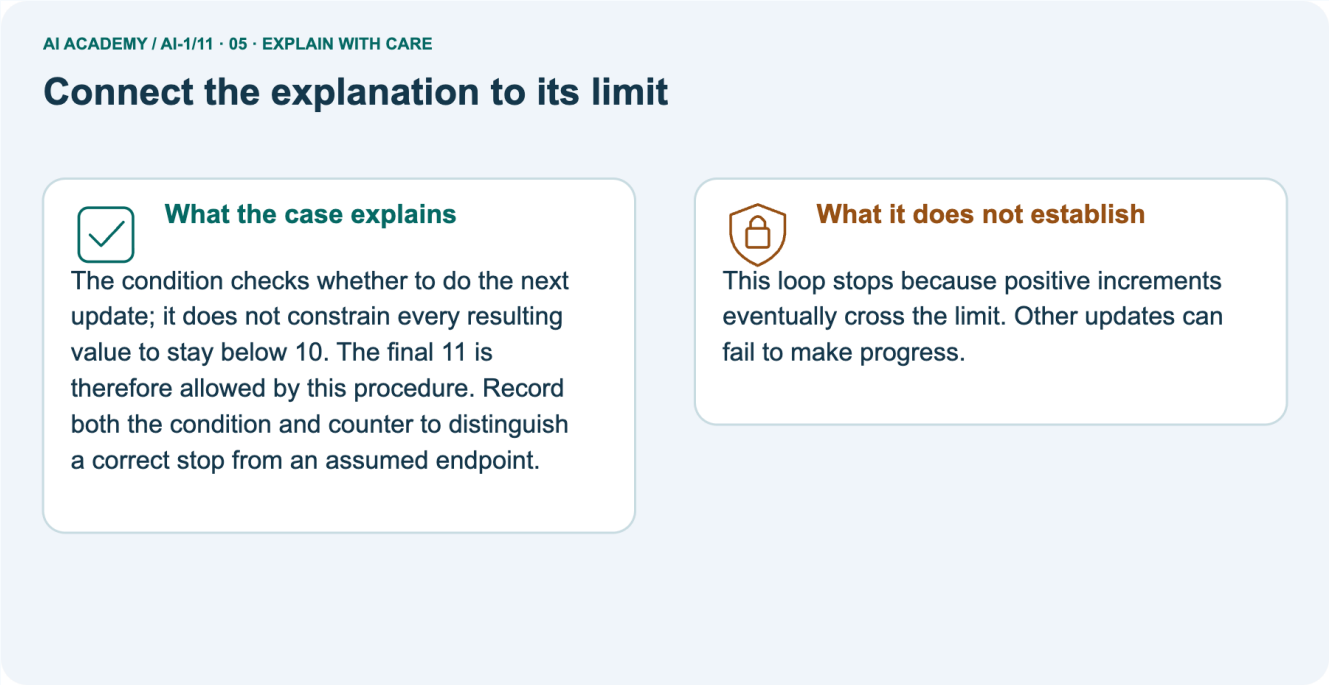
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Final counter 11 after three additions. Changed condition: Start at 10 instead of 2. Expected result: The **loop** runs zero times and reports 10.

Reason: Its first condition is already false, so no addition happens.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



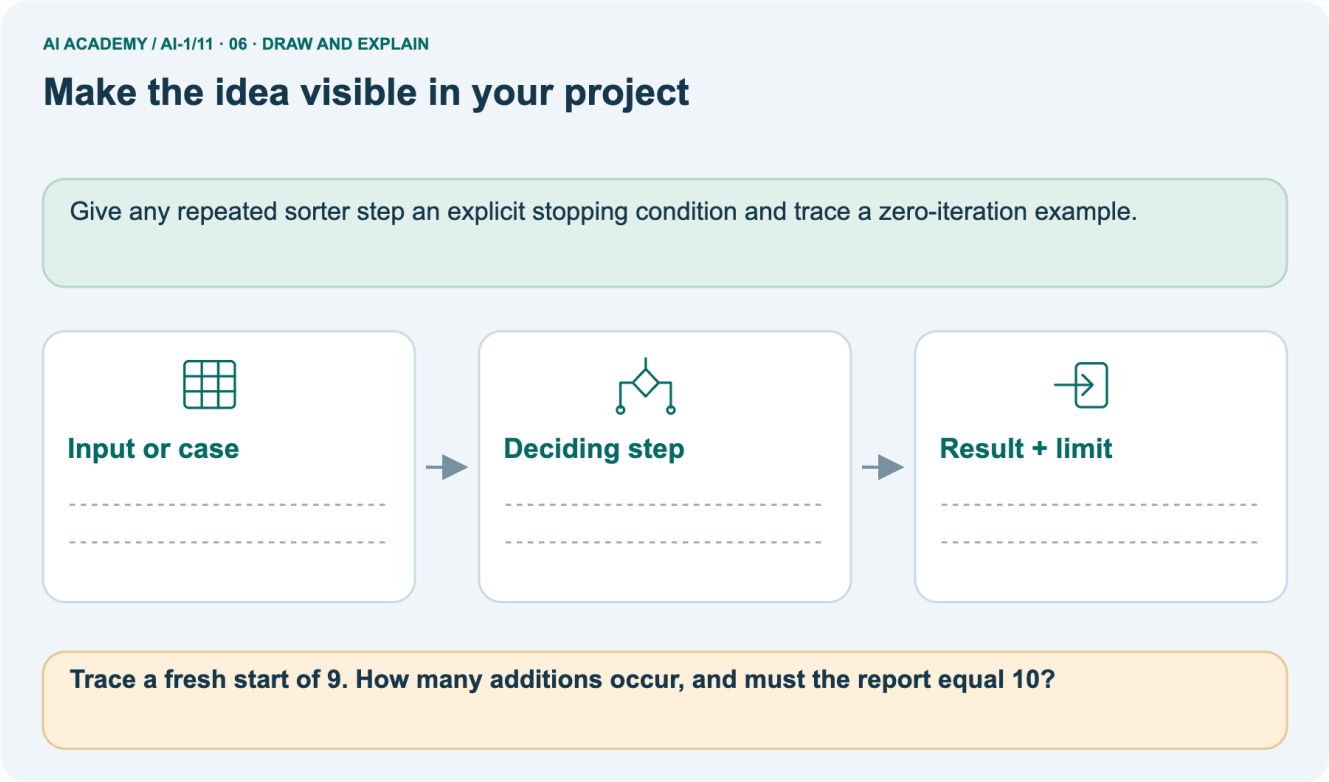
Follow the labelled picture with the matching explanation. Open the vector diagram

The condition checks whether to do the next update; it does not constrain every resulting value to stay below 10. The final 11 is therefore allowed by this procedure.

Record both the condition and counter to distinguish a correct stop from an assumed endpoint. Limit: This **loop** stops because positive increments eventually cross the limit. Other updates can fail to make **progress**.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Give any repeated sorter step an explicit **stopping condition** and trace a zero-**iteration** example. Draw your **input**, deciding step, result and limit. Supported question: Trace a fresh start of 9.

How many additions occur, and must the report equal 10?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.